

G Byte 6

A six-voice lo-fi digital drum machine with the gritty 8, 10 and 12-bit sound of the mid-80s rhythm boxes.



Overview

G Byte 6 is a six-voice drum machine in the spirit of the gritty mid-80s Japanese 12-bit rhythm boxes of the early digital era. There are no samples: every hit is synthesised live and then pushed through a vintage converter, which is where the character comes from.

The six voices are KICK (a sine with a fast pitch drop and an FM thump that dies away), SNARE (an FM tone plus band-passed noise), CL HAT and OP HAT (a cluster of six out-of-tune square waves and noise, high-passed), CLAP (quick noise bursts and a tail) and RIM (an FM ping with a wooden body). The closed hat chokes the open hat, as on a real kit.

Every voice has its own converter. The sound is sampled at the converter RATE with no anti-alias filter, so the folded-back grain is part of the sound, and quantised to 8, 10 or 12 BITS. As on the originals, TUNE moves the voice's converter clock, so a voice tuned down is also grainier. Each converter is followed by a steep output low-pass (FILTER). G Byte 6 has no sequencer: play it from MIDI drum notes.

Reach for it for

- 80s pop, synth-pop and electro funk drum sounds
- Crunchy, lo-fi hip-hop kits
- Bit-crushed, aliased 8-bit percussion
- Layering grit and click under a cleaner drum track
- A light, quick drum machine to play from a MIDI controller or MIDI items

Loading and playing it

1. Insert G Byte 6 on a new track. It is an instrument: it makes its own sound and needs no audio input.
2. Arm the track for recording and set its input to your MIDI keyboard or pads, or put a MIDI item on the track. It plays General MIDI drum notes (see MIDI notes below).
3. Click a pad along the bottom to hear a voice. The LED above the pad lights on every hit, and the MIDI light in MASTER flashes when a note arrives.
4. If you play on a particular MIDI channel, set CHANNEL in MASTER; OMNI listens to all 16.

Controls

The header

PRESETS	Opens the full preset list. The last entry, Reset all to defaults, loads Init and also resets MASTER and IN / OUT.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

LCD

A green two-line dot-matrix display. The top line shows the voice that last played (or the one whose knob you are turning) with its TUNE and DECAY. The bottom line shows the BITS, the converter RATE, the FILTER frequency and the velocity of the last hit (V--- until something has played).

CONVERTER

BITS (8 / 10 / 12)	The converter's depth (12 by default). 12 is clean with a little grit on quiet tails; 10 is grainier; 8 adds clear quantising noise and a crunchy, buzzing decay.
CONVERTER RATE (22K / 26K / 32K / 44K)	The sample clock, about 22, 26, 31 or 44 kHz (32K by default). Lower rates fold more of the high end back as metallic, aliased grain.
FILTER	The 4-pole output low-pass after every converter, 20 - 100% of the converter's Nyquist frequency (80% by default). The knob shows the frequency in kHz. Lower settings give the soft, band-limited top of the period; 100% lets all the grain through.

Voice columns

Six columns with the same four knobs: KICK, SNARE, CL HAT, OP HAT, CLAP and RIM. The MIDI notes for each voice are printed at the top right of its column.

TUNE	Pitch, -12 to +12 semitones (0 by default). It also moves the voice's converter clock, so lower tunings are grainier and duller, and higher ones cleaner.
DECAY	Length, 25 - 400% of the voice's natural decay (100% by default).
LEVEL	The voice's volume, 0 - 100. Defaults: KICK 85, SNARE 75, CL HAT 60, OP HAT 55, CLAP 70, RIM 60.
PAN	Position, L100 to R100. Defaults: KICK and SNARE centre (C), both hats R25, CLAP L15, RIM L30.

Pads

Six pads, one per voice, along the bottom. Click a pad to play its voice at full velocity. The LED above each pad lights on every hit, from the pad or from MIDI.

MASTER

VOLUME	Output level, -24 to +12 dB (0 dB by default).
VELOCITY	How much MIDI velocity changes the level, 0 - 100% (80% by default). At 0 every hit plays at full level.
DRIFT	0 - 100% (15% by default). Being a digital box, G Byte 6 keeps it very subtle: each hit is retuned at random by up to 12 cents at 100% (under 2 cents at the default). Nothing wanders between hits.
CHANNEL	The MIDI channel: OMNI (all channels, the default) or 1 - 16.
MIDI	Flashes when a drum note arrives.
OUT meters	The output level, left and right.

IN / OUT

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises. 0% by default.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). While the output clips, the CLIP key flashes red.
MONO	Makes everything below the set frequency mono, 20 - 500 Hz (fully left = OFF, the default). Use it to keep the low end centred.
WIDTH	Widens the stereo image above the MONO frequency, from +0% up to +100% (twice the side signal).

Knobs: drag up or down (Shift for fine moves); double-click to reset a knob to its default.

MIDI notes

G Byte 6 plays from General MIDI drum notes, with velocity. Other MIDI passes straight through.

KICK	Notes 35, 36
SNARE	Notes 38, 40
CL HAT	Notes 42, 44 (closed and pedal hat). Both choke the open hat.
OP HAT	Note 46
CLAP	Note 39
RIM	Note 37

Presets

Presets set BITS, RATE, FILTER and the six voices. VOLUME, VELOCITY, DRIFT, the MIDI channel and IN / OUT are left as they are, except with Reset all to defaults.

Init	12 bits, 32K, filter 80%, every voice at its defaults.
Classic 80s Pop	12 bits at 32K with a slightly softer filter; a longer snare and clap, tighter closed hat.
Dirty 8-Bit	8 bits at 22K with a dark filter, every voice tuned down a little and lengthened: crunchy and aliased.
Crunchy Hip-Hop	10 bits at 26K and a low filter; a long, low kick and everything tuned down for a dusty, gritty kit.
Synth Pop	12 bits at 44K with the filter almost open; everything tuned up a little, a long snare and clap.

Electro Funk	A long, booming kick tuned down, a short, high snare and a high rim, with tight hats.
Lo-Fi Tape Kit	10 bits at 22K with the darkest filter (42%): soft, muffled and grainy.
Bright 12-Bit	12 bits at 44K with the filter fully open, every voice tuned up: crisp and bright.
Deep Boom Kit	12 bits at 26K; a very long kick tuned down 6 semitones and everything else tuned well down.
Tight & Snappy	12 bits at 32K, every voice tuned up 2 or 3 semitones and shortened to around half length.
Reset all to defaults	Init, plus VOLUME, VELOCITY, DRIFT, CHANNEL and the IN / OUT section back to their defaults.

Tips

- For the most lo-fi sound, combine 8 bits, the 22K rate and a low TUNE: all three add grain.
- Tuning a voice up both raises its pitch and cleans it up; to get a high but grainy voice, lower the RATE as well.
- Long DECAY settings at 8 bits bring out the buzzing quantising noise on the tails - a big part of the period sound.
- Turn FILTER down when the kit sounds harsh; turn it up to 100% to hear all the aliasing.
- Set VELOCITY to 0 for the flat, machine-like dynamics of the original boxes.
- Use MONO at around 120 Hz to keep the kick centred when you widen the kit with WIDTH.

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